

March 2021

DeepFake Generation & Detection

David Weinflash

Research Papers

DeepFake Generation

- *First Order Motion Model for Image Animation*
 - Siarohin *et al.*
 - Neural Information Processing Systems (NeurIPS) 2019

DeepFake Detection

- *Video Face Manipulation Detection Through Ensemble of CNNs*
 - Bonettini *et al.*
 - International Conference on Pattern Recognition (ICPR) 2020

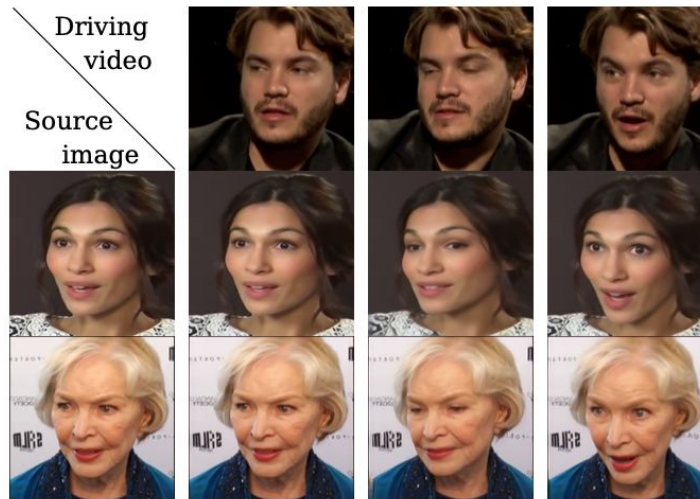
DeepFake Generation

First Order Motion Model for Image Animation

First Order Motion Model for Image Animation

Problem

- **Image Animation:**
 - Automatically synthesize videos by combining the contents of a source image with the motion patterns derived from a driving video.



First Order Motion Model for Image Animation

Problem

- **Image Animation:**
 - Automatically synthesize videos by combining the contents of a source image with the motion patterns derived from a driving video.
- **Challenge:**
 - Traditional approaches (GANs and VAEs) rely on pre-trained models built upon ground-truth data annotations.

First Order Motion Model for Image Animation

Problem

- **Solution:**
 - Introduce a framework that does not depend on prior information or annotated data sets.

First Order Motion Model for Image Animation

Problem

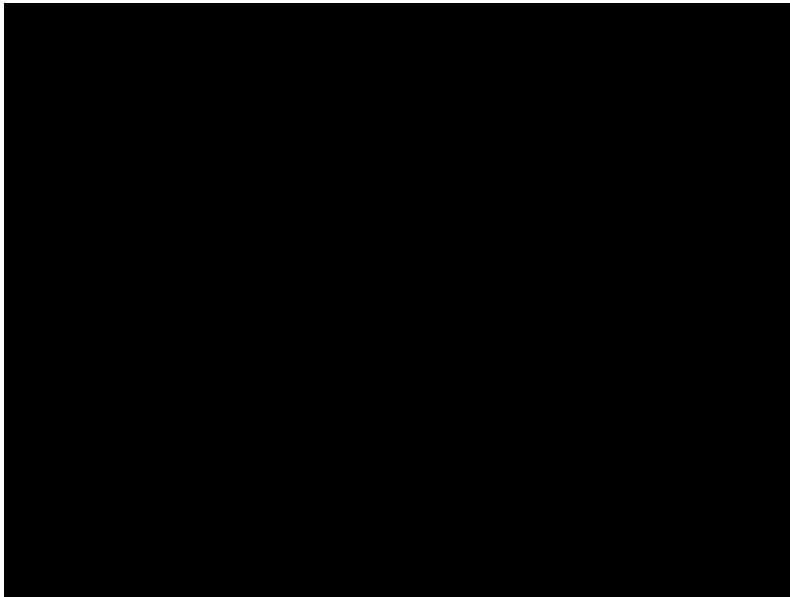
- **Solution:**
 - Introduce a framework that does not depend on prior information or annotated data sets.
- **Framework:**
 - *Motion Estimation Module*
 - Encoder-decoder network predicts a dense motion field aligning the driving video and the source image.
 - *Image Generation Module*
 - Utilize a CNN and output of *motion estimation module* to generate a moving version of the source image.

First Order Motion Model for Image Animation

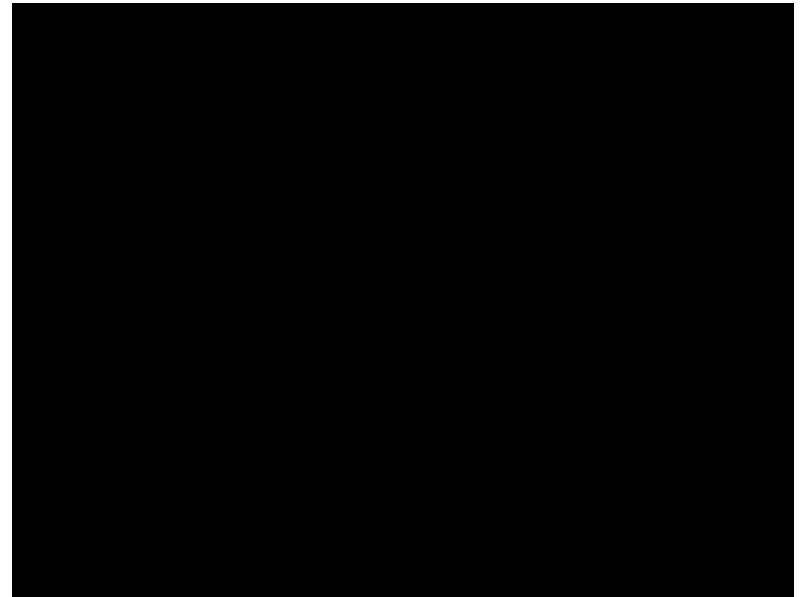
Analysis

- **Claim #1:**
 - *Our method significantly outperforms state-of-the-art image animation methods and can handle high-resolution datasets where other approaches generally fail.*

1 Mbps



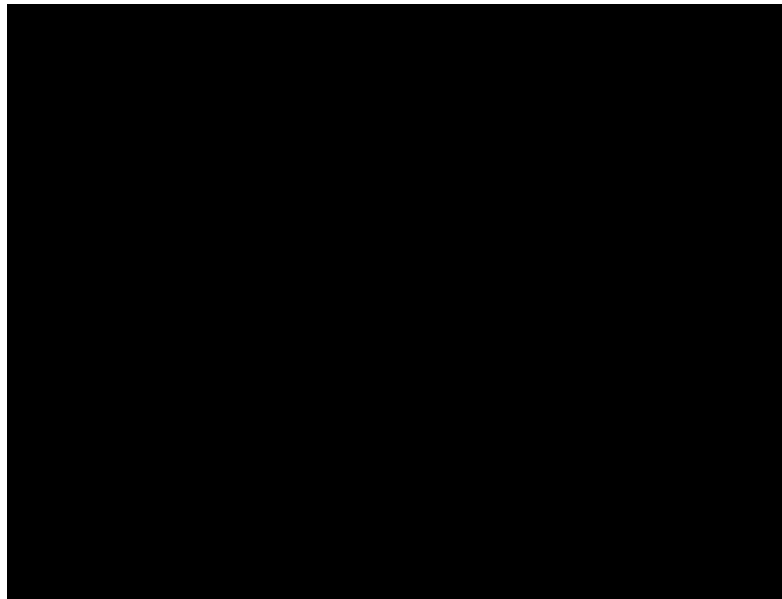
8 Mbps



First Order Motion Model for Image Animation

Analysis

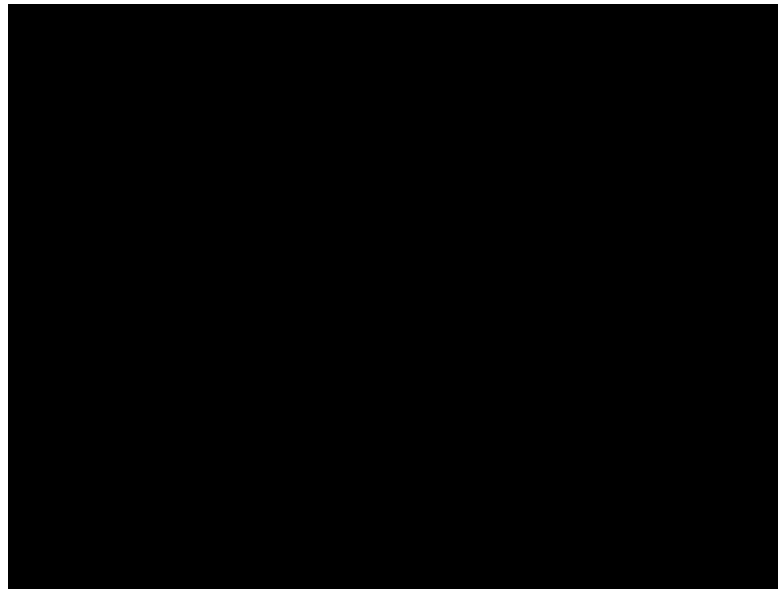
- **Claim #2:**
 - *We introduce an occlusion-aware generator, which adopts an occlusion mask to inpaint object parts that are not visible in the source image and should be inferred from the context.*



First Order Motion Model for Image Animation

Analysis

- **Claim #3:**
 - *One limitation of transferring relative motion is that we need to assume that the objects in the source image and driving video have similar poses.*



DeepFake Detection

Video Face Manipulation Detection Through Ensemble of CNNs

Video Face Manipulation Detection Through Ensemble of CNNs

Problem

- **DeepFake Detection:**
 - Identifying DeepFake videos in real-world scenarios.



Video Face Manipulation Detection Through Ensemble of CNNs

Problem

- **DeepFake Detection:**
 - Identifying DeepFake videos in real-world scenarios.
- **Challenge:**
 - Detecting the ever-growing production of DeepFakes across the internet requires a solution that is robust, efficient and scalable.

Video Face Manipulation Detection Through Ensemble of CNNs

Problem

- **Solution:**
 - Fuse together state-of-the-art CNNs to accurately detect facial manipulation artifacts in a minimal amount of time.

Video Face Manipulation Detection Through Ensemble of CNNs

Problem

- **Solution:**
 - Fuse together state-of-the-art CNNs to accurately detect facial manipulation artifacts in a minimal amount of time.
- **Framework:**
 - *EfficientNet Models*
 - Use attention layers to teach the network what regions of the input image are most important to analyze during classification.
 - Use siamese training to extrapolate additional information from the data and uncover generalizabilities.

Video Face Manipulation Detection Through Ensemble of CNNs

Analysis

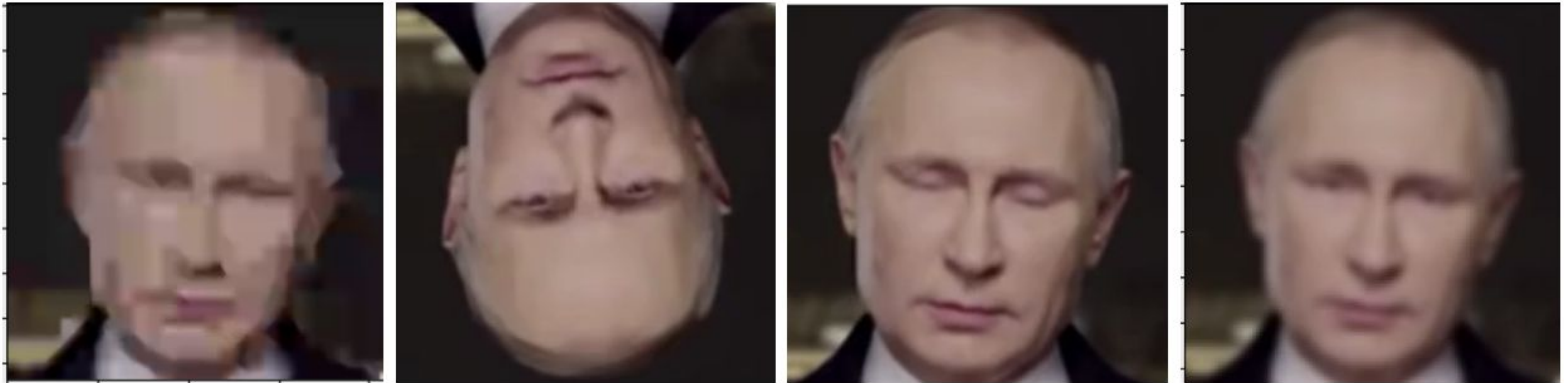
- **Claim #1:**
 - *Network fusion helps both the accuracy of the DeepFake detection and the quality of the detection.*



Video Face Manipulation Detection Through Ensemble of CNNs

Analysis

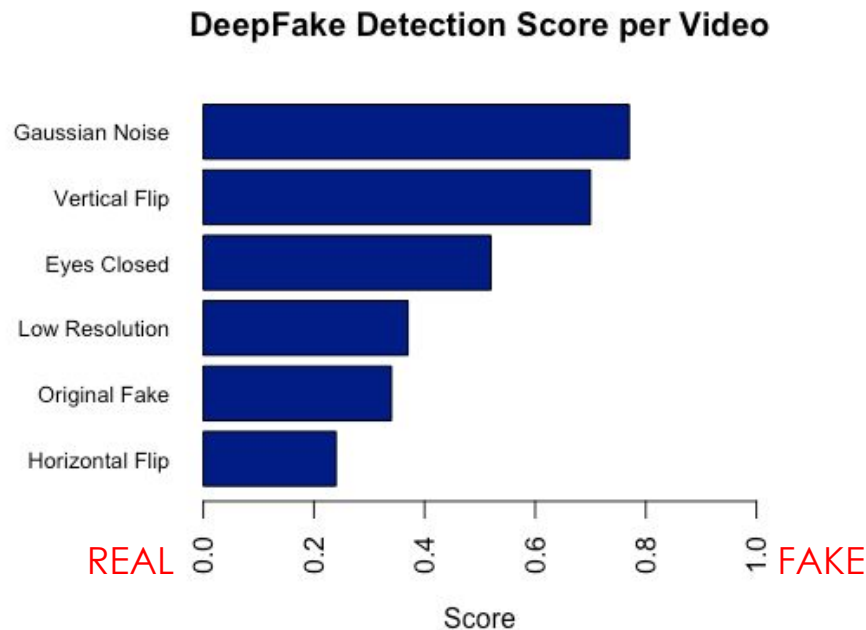
- **Claim #2:**
 - *To make our models more robust, we perform data augmentation operations (downscaling, horizontal flipping, noise addition, etc.) on the input faces.*



Video Face Manipulation Detection Through Ensemble of CNNs

Analysis

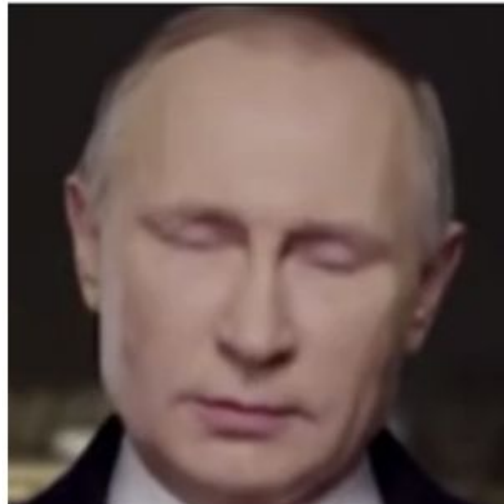
- **Claim #2:**
 - *To make our models more robust, we perform data augmentation operations (downscaling, horizontal flipping, noise addition, etc.) on the input faces.*



Video Face Manipulation Detection Through Ensemble of CNNs

Analysis

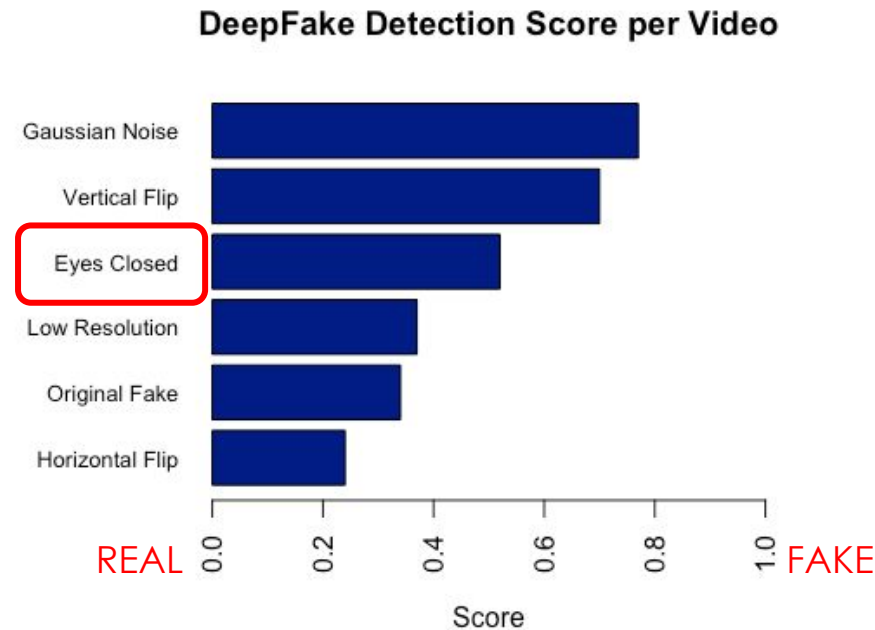
- **Claim #3:**
 - *Roughly modeled eyes and teeth, showing excessively white regions, are the main trademarks of DeepFake generation methods.*



Video Face Manipulation Detection Through Ensemble of CNNs

Analysis

- **Claim #3:**
 - *Roughly modeled eyes and teeth, showing excessively white regions, are the main trademarks of DeepFake generation methods.*



Thank you!